

Effects of temperature on the dynamic characterization of composite steel-concrete highway decks

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Abstract: Variations in experimentally identified modal parameters are commonly used as indicators of the structural condition of bridges, both during acceptance testing and structural health monitoring. However, interpreting these variations becomes complex when environmental effects - particularly temperature - are not explicitly considered, as they can significantly influence the dynamic response of steel structures. This study analyses the impact of daily temperature fluctuations on the natural frequencies of a composite steel-concrete highway deck, utilizing a dynamic testing campaign conducted over the course of a summer day characterized by high temperatures and marked differences between daytime and evening conditions. Twelve consecutive data acquisition sessions were performed using Operational Modal Analysis techniques based on ambient excitation. The results demonstrate a clear sensitivity of the dynamic parameters to temperature, even in the absence of structural changes. The first natural frequency decreases by approximately 13% as the temperature rises from 38°C to 46°C, returning to its initial value during the evening hours. The case study thus highlights the necessity of recording and normalizing thermal conditions when planning and interpreting dynamic tests and monitoring systems for steel and composite decks.

Keywords: Dynamic characterization, steel bridge decks, temperature effects, Operational Modal Analysis, structural health monitoring, thermal influence.

1 Introduction

Dynamic characterization tests are a well-established tool for the comprehensive assessment of the structural behaviour of bridges and viaducts. By identifying natural frequencies, mode shapes, and damping coefficients, it is possible to obtain concise information regarding mass distribution, effective stiffness, and the boundary conditions of the structure. Consequently, modal parameters are frequently used both during acceptance testing - to verify the behaviour predicted in the design - and within structural health monitoring systems aimed at detecting potential changes over time.

For major structures, the Technical Standards for Construction (NTC) highlight the value of supplementing static tests with dynamic tests to compare the experimental fundamental period with the design-predicted value [1]. Similarly, the Guidelines for the

classification and risk management, safety assessment, and monitoring of existing bridges - along with the UNI 10985 standard - emphasize the importance of proper test planning, the selection of the instrumentation layout, and the interpretation of results [2, 3].

However, using natural frequencies as an indicator of structural condition requires careful interpretation of the experimental data rather than a purely automated analysis.

A change in frequency does not unequivocally indicate an alteration in the structure's health, as it may stem from variations in mass, stiffness, or boundary conditions, or from reversible environmental factors. Among the latter, temperature is particularly significant for steel and steel-concrete composite decks, where restrained thermal expansion, the interaction between the slab and steel girders, and changes in contact conditions can alter the system's apparent stiffness. This paper analyzes this issue with reference to the Himera viaduct on the A19 Palermo-Catania motorway, which was the subject of a dynamic characterization campaign conducted over the course of a full summer day. High daytime temperatures and the temperature fluctuation extending into the evening made it possible to observe significant variations in the identified frequencies on the same structure, without any permanent modifications having been made.

The case study thus offers practical insights for interpreting thermal effects during dynamic testing and for ensuring the repeatability of comparisons over time.

2 Natural frequencies, stiffness, and environmental variability

In a linear dynamic system, natural frequencies depend on the mass distribution and stiffness. In a simplified form, for a single-degree-of-freedom oscillator, this dependence can be expressed by Eq. (1).

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (1)$$

where k represents the equivalent stiffness and m the participating mass. Although real structures are continuous systems with multiple degrees of freedom, Eq. (1) highlights a fundamental concept: for a given mass, a reduction in effective stiffness leads to a reduction in natural frequencies. For this reason, in structural monitoring, a drop in frequency is often interpreted as a possible sign of deterioration or damage.

This interpretation is valid only when the comparison is made under identical boundary and operating conditions. Thermal variations can cause differential expansion, modify the degree of restraint, alter steel-concrete interaction, and induce reversible strain or stress states. In composite decks, these effects can be particularly pronounced, as materials with different thermal and inertial behaviours contribute jointly to the structural response.

It follows that dynamic testing retains high diagnostic value only if modal data are interpreted in conjunction with environmental conditions. Consequently, during acceptance testing, periodic inspections, and Structural Health Monitoring operations, temperature must be measured, recorded, and - whenever possible - used to normalize

dynamic parameters before comparing them against thresholds or reference values [6, 7].

Experimental literature has extensively documented this issue. Annual monitoring of the Z24 Bridge highlighted the need to model environmental variability - specifically thermal effects - to distinguish normal frequency fluctuations from damage events [8]. Analyses of the Ting Kau Bridge revealed correlations (including non-linear ones) between structural temperatures and modal parameters identified through long-term monitoring [9]. With specific regard to steel decks, Ding and Li observed a general decrease in frequencies as temperature rose, as well as varying sensitivity among individual modes [10]. This study is situated within this context, focusing on the intraday frequency variability of a composite steel-concrete highway deck through twelve consecutive OMA acquisitions.

3 Case study: the Himera viaduct

The Himera viaduct is located along the A19 Palermo-Catania motorway, near the Scilato interchange, on the carriageway heading towards Catania. The new deck replaces the previous viaduct and consists of three continuous spans using a composite steel-concrete system. The structural configuration features two main steel girders with variable cross-sections, connected transversely by vertical and horizontal bracing, and a reinforced concrete deck slab.

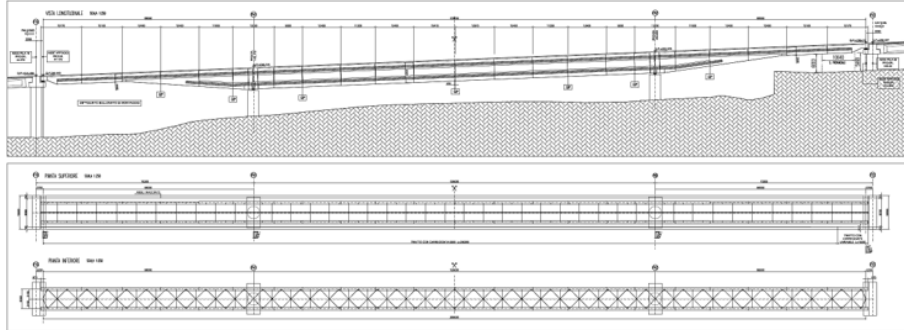


Fig. 1. Plan and longitudinal profile of the Himera viaduct



Fig. 2. General view of the motorway deck during the testing campaign

The two side spans have a span length of 68.00 m, measured between the bearing axes, while the central span has a length of 129.63 m. The roadway is 9.30 m wide, with 0.75 m edge curbs; the spacing between the two main girders is 6.30 m. The girder depth varies from a minimum of 2.50 m to a maximum of 6.50 m. These characteristics make the structure particularly suitable for assessing the dynamic sensitivity of a long-span composite deck, in which the steel component, the concrete slab, and the boundary conditions all contribute to defining the overall stiffness.

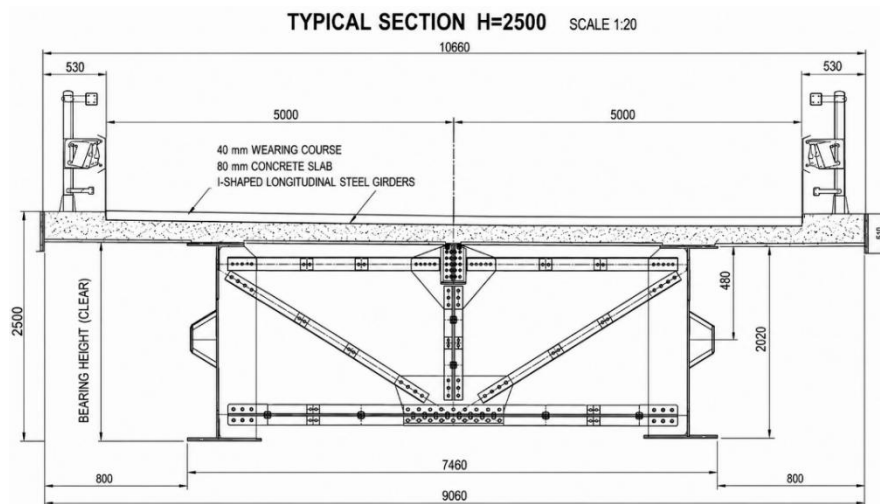


Fig. 3. Cross-section of the bridge deck

4 Experimental campaign

The testing campaign was conducted on July 30, 2020, through dynamic ambient data acquisition in the absence of vehicular traffic. Structural excitation relied on ambient forces and natural vibrations present during the test, following an approach consistent with Operational Modal Analysis techniques [4, 5].

Twelve consecutive acquisitions were performed, each lasting 2,700 seconds (45 minutes). The first measurement began at 11:04 and the last concluded at 23:16, allowing the structural response to be monitored from the hottest daytime hours through to evening conditions.



Fig. 4. Timeline of the dynamic acquisitions carried out during the testing day

The transition from the first to the second setup took place between the sixth and seventh data acquisition runs. Three reference accelerometers were kept in the same positions to link the data from the two layouts and reconstruct the overall behaviour of the entire deck.

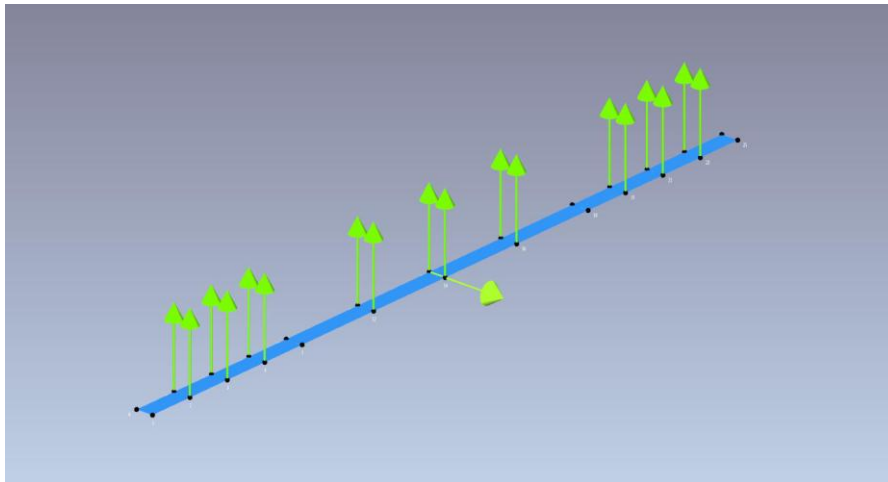


Fig. 5. Overall layout of the accelerometers after merging the two test setups

The main acquisition parameters remained constant throughout all measurements: ambient excitation in the absence of vehicular traffic, two experimental setups, ten vertical channels and one transverse horizontal channel per setup, a sampling frequency of 100 Hz, and an acquisition duration of 2700 s. Measurement points were located at significant sections - such as quarter-spans and mid-spans - to reconstruct the primary mode shapes. Signal processing enabled the identification of natural frequencies, mode shapes, and damping ratios.

5 Modal identification and experimental results

Analysis of the acquired signals made it possible to identify the deck's principal natural frequencies. The mean values derived from the entire set of acquisitions were 1.16, 1.84, 2.10, 2.49, 3.16, and 4.02 Hz. The average damping factor, estimated in the time domain using the logarithmic decrement method on signal windows characterized by sufficiently distinct oscillations, was found to be in the order of 1.7%.

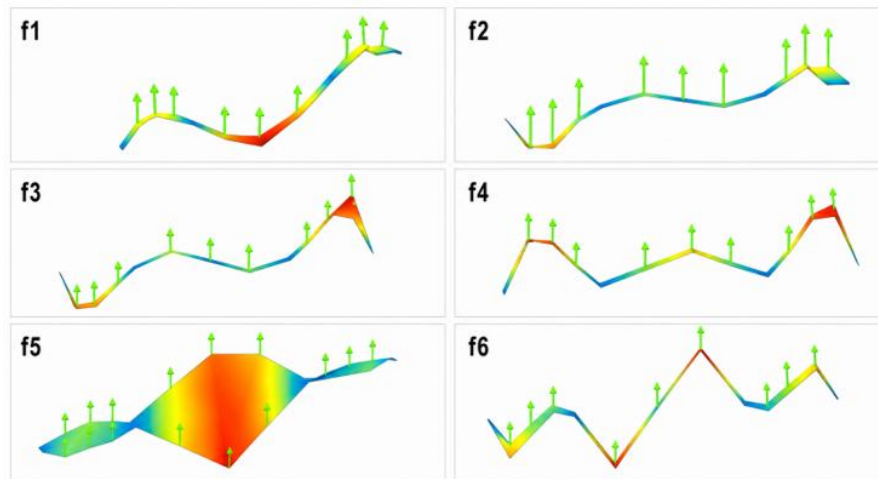


Fig. 6. Main modal shapes identified on the deck and associated average frequencies

The identified frequencies provide a comprehensive description of the deck's response. However, using them as reference values for tests conducted at different times requires taking into account changes in environmental conditions. In the case at hand, the test day was characterized by high temperatures and significant differences between midday and evening hours.

6 Effects of temperature on the identified frequencies

The influence of temperature on the dynamic response of a bridge deck cannot be attributed solely to variations in the mechanical properties of the materials. In composite

steel-concrete structures, thermal effects operate at multiple levels: they induce longitudinal expansion, can generate thermal gradients between the slab and the steel girders, alter the deck's deformation state, and may temporarily modify contact and boundary conditions at the supports and connection devices. From a dynamic perspective, these phenomena result in a change in the effective stiffness experienced by the structure during vibration. Since natural frequencies depend on the ratio of stiffness to mass, even reversible thermal fluctuations can produce measurable shifts in modal parameters, despite the absence of permanent changes to the structural state.

The effect of temperature does not necessarily manifest with the same intensity across all vibration modes. Each mode engages different portions of the structure and is affected differently by stiffness distribution, boundary conditions, and the interaction between primary and secondary elements. Consequently, the analysis must critically evaluate the sensitivity of individual modes to thermal conditions, rather than assuming a uniform shift in frequencies.

To highlight the influence of temperature, three datasets representative of the daily thermal cycle were selected: Acquisition 1, recorded between 11:04 and 11:49 with an external temperature of 38°C; Acquisition 4, recorded between 14:18 and 15:03 with a temperature of 46°C; and Acquisition 12, recorded between 22:31 and 23:16 with a temperature of 33°C. The comparison is particularly significant because it involves the same structure over the course of a few hours, during which it is not reasonable to assume permanent structural changes.

Table 1. Comparison of representative acquisitions at different temperatures

Acq.	Time	T [°C]	f1 [Hz]	f2 [Hz]	f3 [Hz]	f4 [Hz]	f5 [Hz]	f6 [Hz]
1	11:04-11:49	38	1,16	1,81	2,05	2,49	3,18	4,03
4	14:18-15:03	46	1,01	1,75	2,01	2,29	3,02	3,97
12	22:31-23:16	33	1,17	1,83	2,12	2,49	3,13	4,03

The comparison reveals a reduction in frequencies during the hottest hours, followed by a recovery in the evening. Specifically, the first frequency drops from 1.16 Hz in the first acquisition to 1.01 Hz in the fourth—a reduction of approximately 13%—before rising back to 1.17 Hz in the twelfth acquisition. Similar behavior is observed for other modes; for instance, the fourth frequency shifts from 2.49 Hz to 2.29 Hz and subsequently returns to 2.49 Hz.

The reversibility of this phenomenon is the most significant aspect from an interpretive standpoint. Since the variation was observed in the same structure over the course of just a few hours, it cannot reasonably be attributed to the progression of damage or a permanent structural alteration. Consequently, the experimental data must be interpreted as the result of varying thermal conditions, which temporarily alter the apparent stiffness of the deck and, in turn, the identified frequencies.

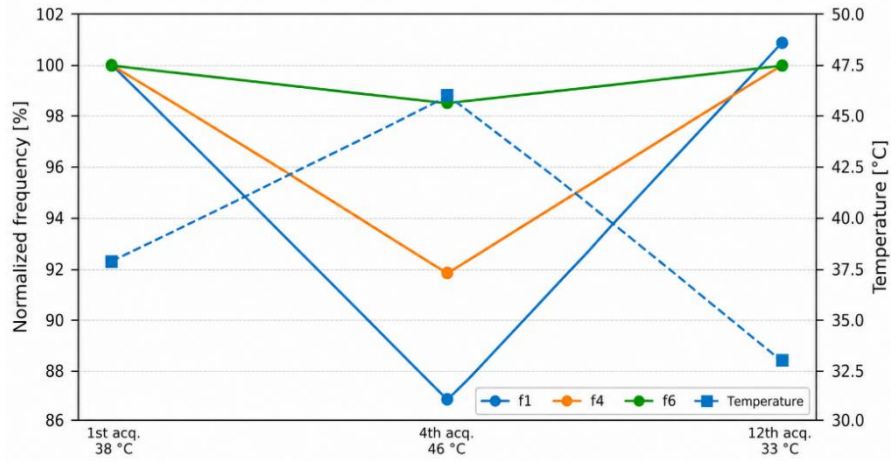


Fig. 7. Evolution of external temperature and normalised frequencies in three representative acquisitions

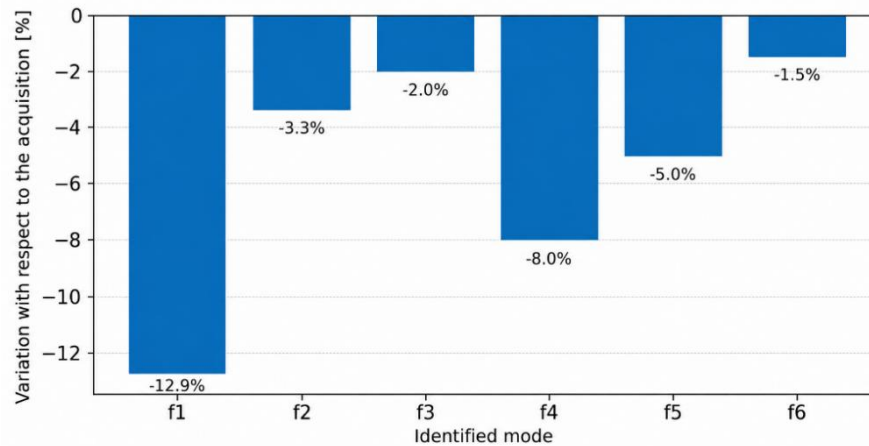


Fig. 8. Percentage variation of the frequencies in the fourth acquisition with respect to the first acquisition

To allow for a direct comparison of the results, Fig. 7 presents frequencies normalized against Acquisition 1, while Fig. 8 quantifies the reductions observed at the peak representative temperature. All selected modes exhibit a decrease during Acquisition 4 followed by a recovery in the evening hours, albeit with varying magnitudes. The first mode proves to be the most sensitive, showing a 12.9% reduction, followed by the fourth and fifth modes with variations of 8.0% and 5.0%, respectively. Variations in the other modes are more limited. This non-uniform response suggests that the thermal effect depends on the varying degrees to which global and local stiffnesses, boundary conditions, and steel-concrete interaction contribute to each mode shape.

7 Discussion

The case of the Himera viaduct clearly demonstrates the role of environmental conditions in the interpretation of modal parameters. Frequency variations were observed on the same structure, on the same day, and without any intervening structural modifications between tests. Temperature thus acts as a contextual variable capable of significantly altering the dynamic response of the bridge deck.

This finding is particularly significant in the context of dynamic acceptance testing and structural health monitoring. A reduction in natural frequencies could be interpreted as a loss of stiffness associated with deterioration, damage, or changes in boundary conditions. However, without measuring or correcting for thermal conditions, such an interpretation could be misleading. Although dynamic data are highly informative, they should not be viewed as absolute indicators independent of the operational context.

From a practical standpoint, comparing dynamic tests conducted at different times should involve planning to replicate climatic conditions as closely as possible. When this is not feasible, it is necessary to record air temperature—and preferably the surface temperature of structural elements—to assess the potential normalization of modal parameters. This requirement is especially important for steel or composite steel-concrete decks, which can be highly sensitive to daily temperature fluctuations.

The available dataset is limited to a single summer day and to measurements of external air temperature. Consequently, the results do not allow for the development of a general predictive model for the temperature-frequency relationship, nor do they enable the separation of effects attributable to uniform temperature, thermal gradients, and bearing conditions. Long-term monitoring, supplemented by sensors distributed across the deck, the steel beams, and at the supports, would allow for a more comprehensive characterization of environmental variability and its comparison with any changes in structural condition.

8 Conclusions

This study analysed the influence of thermal conditions on the dynamic characterization of a continuous composite steel-concrete highway deck, using the Himera viaduct on the A19 Palermo-Catania motorway as a case study. The experimental campaign, conducted over the course of a full summer day involving twelve sets of ambient dynamic measurements, made it possible to observe the evolution of modal parameters in relation to temperature.

A comparison of representative measurements revealed a significant reduction in frequencies during the hottest period, followed by a recovery during the evening hours. The first natural frequency dropped by approximately 13% between measurements taken at 38°C and 46°C, subsequently returning to its initial value when the temperature fell to 33°C. Other modes also exhibited reversible variations, albeit with varying degrees of sensitivity.

Since these variations were observed on the same structure within a limited timeframe, they must be interpreted as a reversible environmental effect rather than an

indicator of permanent structural changes. Consequently, dynamic testing must be accompanied by accurate monitoring of thermal conditions, and longitudinal comparisons should be performed under comparable climatic conditions or by employing appropriate environmental normalization procedures.

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